



# COMPLETE DEEP-DIVE RESEARCH REPORT: STERLING E-MOBILITY SOLUTIONS LIMITED

## EXECUTIVE SUMMARY

Sterling E-Mobility Solutions Limited (SEM) — formerly Sterling Gtake E-Mobility — is India's largest manufacturer of Motor Control Units (MCUs) for electric vehicles, a 100% subsidiary of Sterling Tools Limited (BSE/NSE listed). Incorporated on **January 10, 2020**, SEM has rapidly evolved from a single-product MCU supplier into a comprehensive EV powertrain and power electronics solutions provider with **720,000 units/year** manufacturing capacity and over **50% market share** in India's MCU segment. <sup>[1] [2] [3]</sup>

## SECTION 1: COMPANY ORIGIN

### Founding Details

| Attribute                   | Information                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------|
| Exact Date of Incorporation | January 10, 2020 <sup>[2] [4]</sup>                                                    |
| CIN                         | U31909DL2020PLC360123 <sup>[2]</sup>                                                   |
| Original Name               | Sterling Gtake E-Mobility Limited <sup>[5] [6]</sup>                                   |
| Rebranding Date             | October 2025 (renamed to Sterling E-Mobility Solutions Limited) <sup>[5] [7]</sup>     |
| Parent Company              | Sterling Tools Limited (established 1979) <sup>[5] [11]</sup>                          |
| Original Promoters          | Atul Aggarwal (MD), Anish Agarwal (Director), Sterling Group <sup>[2]</sup>            |
| Current Managing Director   | Atul Aggarwal (MBA from Kenan-Flagler, <u>B.Com</u> from SRCC) <sup>[8] [9]</sup>      |
| Director                    | Jaideep Wadhwa (appointed Sept 2022 as Director, became MD of SEM) <sup>[10] [2]</sup> |

### Why the Company Was Created

Sterling Tools Limited established an **E-Mobility Division** in **2018** with the objective of supporting Auto OEMs with leading-edge technology products in automotive electronics and electric vehicle domains. The subsidiary was incorporated in 2020 to: <sup>[11] [3]</sup>

1. Capitalize on India's EV transition under FAME II and growing EV adoption
2. Leverage Sterling Group's 40+ years of automotive component manufacturing expertise <sup>[12]</sup>

- 3. Address the MCU supply gap — India lacked domestic MCU manufacturers at scale
- 4. Enable import substitution through technology partnership with Jiangsu Gtake Electric (China) [7] [1]

Original Vision

"We aim to be India's leading manufacturer and supplier of Electric Vehicle Components & Solutions"[1]

**Mission:** To be an **Engineering-led Company** providing Innovative Solutions as well as best-in-class Application & After-Sales Support to the EV Industry [1]

Original Business Model

- license-based manufacturing of Motor Control Units from Jiangsu Gtake Electric Co. Ltd. (China)
- B2B supply to EV OEMs across 2W, 3W, LCV, and commercial vehicle segments
- Application support + vehicle integration as differentiator [3] [1]

First Product Launched

**Motor Control Units (MCUs)** — the company's inaugural product, covering low voltage (48V) to high voltage (700V) applications [13] [1]

Chronological Timeline: Day 1 to Present

| Year         | Event                                                                          | Business Impact                           |
|--------------|--------------------------------------------------------------------------------|-------------------------------------------|
| 2018         | Sterling Tools establishes E-Mobility Division                                 | Strategic entry into EV domain [11] [3]   |
| Jan 10, 2020 | Company incorporated as "Sterling Gtake E-Mobility Ltd"                        | Legal entity formed for EV components [2] |
| 2020         | Started MCU manufacturing with Jiangsu Gtake license                           | Entered EV components market [1]          |
| 2021         | Manufactured over 300,000 MCUs; supplied to 15+ EV OEMs                        | Achieved scale quickly [14]               |
| FY2022       | Q4 revenue: ₹309 million; monthly run rate ₹100 million                        | Established profitable operations [13]    |
| FY2023       | Expanded to 3W, motorcycle, LCV customers                                      | Product portfolio diversified [13]        |
| FY2024       | EV business grew 90%; EV control unit revenue ₹325 crore                       | Major growth milestone [15]               |
| 2024         | Received "Delivery Performance Runner-Up" award from Ola Electric              | Key customer recognition [1]              |
| 2023         | "Most Trusted Partner" award from Greaves Electric Mobility                    | Industry validation [1]                   |
| FY2025       | Sterling Gtake revenue ₹345.9 crore (9 months); +56% YoY                       | Sustained growth trajectory [16]          |
| FY2025       | Sterling Tools consolidated revenue ₹1,038 crore (+10.6% YoY); PAT ₹58.3 crore | Parent company strong performance [17]    |

| Year     | Event                                                                          | Business Impact                                           |
|----------|--------------------------------------------------------------------------------|-----------------------------------------------------------|
| 2025     | Partnership with <b>Advanced Electric Machines (UK)</b> for magnet-free motors | Technology expansion <sup>[7]</sup>                       |
| 2025     | Partnership with <b>Landworld Technology (China)</b> for OBC/DC-DC converters  | Entered charger segment <sup>[7]</sup> <sup>[18]</sup>    |
| Oct 2025 | Rebranded to "Sterling E-Mobility Solutions Limited"                           | Reflects expanded portfolio <sup>[5]</sup> <sup>[7]</sup> |
| 2025     | Bronze recognition at ACMA Excellence Awards (Localization category)           | Industry acclaim <sup>[1]</sup>                           |
| 2025     | Installed capacity: 720K MCUs/year                                             | Market leadership confirmed <sup>[1]</sup>                |

## SECTION 2: COMPANY EVOLUTION

### Business Transformation

| Phase     | Focus                    | Key Changes                                              |
|-----------|--------------------------|----------------------------------------------------------|
| 2020–2021 | MCU-only supplier        | Single product, licensing model                          |
| 2022–2023 | MCU scale-up             | 300K+ units, 15+ OEMs                                    |
| 2024–2025 | Full powertrain provider | Added motors, chargers, DC-DC converters                 |
| 2025+     | Systems supplier         | Targeting 70% integrated systems by FY28 <sup>[19]</sup> |

### Key Turning Points

1. **2018 Division Creation** — Strategic pivot toward EV before subsidiary incorporation
2. **2020 Incorporation** — Dedicated EV entity with Gtake partnership
3. **2021 Scale Achievement** — 300K MCUs manufactured validated the business model
4. **Ola Electric Partnership** — Became largest customer, driving 90% EV growth <sup>[15]</sup> <sup>[20]</sup>
5. **2025 Technology Partnerships** — AEM (UK) for magnet-free motors; Landworld for OBC/DC-DC
6. **FY28 Systems Target** — 70% integrated systems strategy announced <sup>[19]</sup>

### Major Successes

- **Market Leadership:** India's largest MCU manufacturer with 50%+ market share <sup>[1]</sup>
- **Rapid Scale:** 720K units/year capacity within 5 years
- **Customer Base:** 15+ EV OEMs including Ola Electric, Greaves Electric Mobility
- **90% EV Business Growth** in FY24 <sup>[15]</sup>
- **ACMA Bronze Award 2025** for Localization <sup>[1]</sup>

## Major Challenges

- **Ola Electric Dependency:** Q4 FY25 slowdown after Ola shifted to in-house MCU production for Gen3 models <sup>[17]</sup>
- **Customer Concentration Risk:** Ola Electric was largest customer <sup>[20]</sup>
- **Technology Licensing:** Dependence on Chinese partners (Gtake, Landworld) for core technology

## Strategic Pivots

| Pivot                      | Description                                                                           |
|----------------------------|---------------------------------------------------------------------------------------|
| Product Expansion          | MCU → Motors → Chargers → DC-DC → 3-in-1 integrated systems                           |
| Systems Supplier           | Targeting 70% integrated powertrain systems by FY28 <sup>[19]</sup>                   |
| Localization               | FAME II compliance; ACMA award for NPDD & Localization <sup>[1]</sup> <sup>[13]</sup> |
| Technology Diversification | Magnet-free motors via AEM UK partnership (no rare-earth magnets) <sup>[7]</sup>      |

# SECTION 3: COMPLETE BUSINESS MODEL

## What Sterling E-Mobility Does

SEM is a **Powertrain and Power Electronics Solution Provider** to the EV industry, manufacturing:

| Category           | Products                                                                                     |
|--------------------|----------------------------------------------------------------------------------------------|
| Motor Control      | Motor Control Units (MCUs) — 48V to 700V, 1kW to 200kW <sup>[11]</sup> <sup>[13]</sup>       |
| Motors             | BLDC, PMSM, Induction Motors; Magnet-free motors (via AEM UK) <sup>[21]</sup> <sup>[7]</sup> |
| Charging           | On-Board Chargers (6.6kW, 11kW, 22kW); Off-Board Chargers <sup>[1]</sup> <sup>[12]</sup>     |
| Power Conversion   | DC-DC Converters (2.5kW/6kW) <sup>[12]</sup>                                                 |
| Integrated Systems | 2-in-1 (OBC + DC-DC); 3-in-1 (OBC + DC-DC + PDU) <sup>[12]</sup>                             |
| Power Distribution | PDUs (Power Distribution Units) <sup>[18]</sup>                                              |

## Industries Served

- **2-Wheelers** (electric scooters, motorcycles)
- **3-Wheelers** (E-rickshaws, E-CNGONs)
- **Light Commercial Vehicles** (LCVs above 2.5 tonnes)
- **Passenger Cars** (BEVs)
- **Commercial Vehicles**

## EV Ecosystem Involvement

| Area                    | Involvement Level                                                                   |
|-------------------------|-------------------------------------------------------------------------------------|
| Battery Technology      | Indirect (DC-DC converters interface with batteries)                                |
| Charging Infrastructure | Yes — On-Board Chargers (6.6–22kW), Off-Board Chargers <sup>[1]</sup>               |
| Manufacturing           | Yes — Faridabad plant, 720K MCUs/year capacity <sup>[1]</sup>                       |
| Software                | Yes — Software customization, UDS development, CAN-BUS communication <sup>[1]</sup> |
| Engineering             | Yes — Design, vehicle integration, testing, validation <sup>[1]</sup>               |
| Mobility Solutions      | Yes — Full powertrain solutions for OEMs <sup>[1]</sup>                             |

## Revenue Streams

| Stream                | Description                              | FY24 Contribution                                         |
|-----------------------|------------------------------------------|-----------------------------------------------------------|
| MCU Sales             | Motor Control Units to OEMs              | ~₹325 crore (EV control units) <sup>[15]</sup>            |
| Motor Sales           | Traction motors (BLDC, PMSM, IM)         | Growing segment <sup>[1]</sup>                            |
| Charger Sales         | OBC (6.6–22kW), Off-Board                | New revenue from 2025 <sup>[12]</sup>                     |
| DC-DC Converter Sales | 2.5kW/6kW converters                     | New revenue from 2025 <sup>[12]</sup>                     |
| Integrated Systems    | 2-in-1, 3-in-1 packages                  | Future growth driver (70% target by FY28) <sup>[19]</sup> |
| Application Support   | Vehicle integration, testing, validation | Differentiator, not separately billed <sup>[1] [3]</sup>  |

**Total Parent Company Revenue (FY25):** ₹1,038 crore consolidated; EV business ~42% of mix in H1 FY25 <sup>[22] [17]</sup>

# SECTION 4: PRODUCTS AND TECHNOLOGY

## Complete Product Portfolio

### 1. Motor Control Units (MCUs) — Flagship Product

| Attribute             | Details                                                      |
|-----------------------|--------------------------------------------------------------|
| Purpose               | Control electric motor operation (torque, speed, efficiency) |
| Voltage Range         | 48V to 700V <sup>[13]</sup>                                  |
| Power Range           | 1kW to 200kW <sup>[13]</sup>                                 |
| Motor Types Supported | BLDC, PMSM, Induction Motors <sup>[21]</sup>                 |
| Target Market         | 2W, 3W, LCV, PCV, Commercial EVs                             |
| Capacity              | 720,000 units/year <sup>[1]</sup>                            |
| Market Share          | 50%+ in India <sup>[1]</sup>                                 |

| Attribute             | Details                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Competitive Advantage | India's largest MCU manufacturer; extensive portfolio; application support <a href="#">[1]</a> <a href="#">[3]</a> |
| FAME II Compliance    | Partially localized; more in progress <a href="#">[13]</a>                                                         |

## 2. Traction Motors

| Attribute             | Details                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------|
| Motor Types           | Brushless DC (BLDC), Permanent Magnet Synchronous (PMSM), Induction (IM) <a href="#">[21]</a> |
| Magnet-Free Option    | Yes — via Advanced Electric Machines (UK) partnership <a href="#">[7]</a>                     |
| Purpose               | Provide propulsion for EVs                                                                    |
| Target Market         | 2W, 3W, LCV segments                                                                          |
| Competitive Advantage | Magnet-free motors eliminate rare-earth dependency                                            |

## 3. On-Board Chargers (OBC)

| Product   | Power | Voltage Range                        | Efficiency    | Cooling       | Target                                       |
|-----------|-------|--------------------------------------|---------------|---------------|----------------------------------------------|
| 6.6kW OBC | 6.6kW | Passenger BEV                        | Not specified | Not specified | Passenger cars <a href="#">[12]</a>          |
| 11kW OBC  | 11kW  | LCV >2.5T, BEV                       | Not specified | Not specified | LCV, Passenger cars <a href="#">[12]</a>     |
| 22kW OBC  | 22kW  | 200–450Vdc / 400–650Vdc / 450–750Vdc | ≥94%          | Liquid cooled | High-power applications <a href="#">[12]</a> |

## 4. DC-DC Converters

| Product         | Power        | Target                                    |
|-----------------|--------------|-------------------------------------------|
| 2.5kW/6kW DC-DC | 2.5kW or 6kW | Passenger cars, LCVs <a href="#">[12]</a> |

## 5. Integrated Systems

| System | Components                         | Purpose                                             |
|--------|------------------------------------|-----------------------------------------------------|
| 2-in-1 | 6.6kW/11kW OBC + 2.5kW DC-DC       | Compact power module <a href="#">[12]</a>           |
| 3-in-1 | 6.6kW/11kW OBC + 2.5kW DC-DC + PDU | Integrated powertrain solution <a href="#">[12]</a> |

## 6. Off-Board Chargers

- Wall-mounted chargers for EV charging stations [\[12\]](#)

## Technological Innovations

| Innovation             | Description                                                   | Strategic Value                                                        |
|------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|
| Magnet-Free Motors     | Partnership with AEM UK for motors without rare-earth magnets | Eliminates supply chain risk; cost reduction <sup>[7]</sup>            |
| Localization           | FAME II-compliant MCUs; ACMA award for NPDD & Localization    | Import substitution; government support <sup>[1]</sup> <sup>[13]</sup> |
| Dyno Testing           | Up to ____ kW/1200V testing capability <sup>[1]</sup>         | In-house validation reduces time-to-market                             |
| Software Customization | UDS development, CAN-BUS communication                        | Tailored OEM solutions <sup>[1]</sup>                                  |

## Past Products

- Initial MCU portfolio (2020–2021) — low voltage only

## Upcoming Products

- Multi-function drive units integrating charger, converter, and PDU <sup>[19]</sup>
- Expanded magnet-free motor portfolio via AEM UK <sup>[7]</sup>
- Higher voltage systems for 800V architectures (implied by 750Vdc OBC) <sup>[12]</sup>

# SECTION 5: MANUFACTURING AND OPERATIONS

## Manufacturing Facilities

| Facility            | Location                                                                                  | Purpose                               |
|---------------------|-------------------------------------------------------------------------------------------|---------------------------------------|
| Manufacturing Plant | 12/2, NH-19, Sector 27, Faridabad, Haryana 121003 <sup>[23]</sup>                         | MCU, motor, charger production        |
| Technical Center    | 73, 74, 113, 114, ND Fusion Mall, 1st Phase, BTM Layout, Bangalore 560076 <sup>[23]</sup> | R&D, engineering, application support |

EV Campus: Faridabad facility referred to as "EV campus" where OBC/DC-DC will be produced <sup>[18]</sup>

## Production Capacity

| Product        | Capacity                                         |
|----------------|--------------------------------------------------|
| MCUs           | 720,000 units/year (expandable) <sup>[1]</sup>   |
| Total Strength | 51–200 employees <sup>[11]</sup> <sup>[24]</sup> |

## Supply Chain Structure

| Component            | Partner                                  | Location                                       |
|----------------------|------------------------------------------|------------------------------------------------|
| MCU Technology       | Jiangsu Gtake Electric Co. Ltd.          | Shenzhen, China <sup>[1]</sup>                 |
| Magnet-Free Motors   | Advanced Electric Machines (AEM)         | Washington, UK <sup>[1]</sup>                  |
| OBC/DC-DC Technology | Landworld Technology Co. Ltd.            | Shenzhen, China <sup>[1]</sup> <sup>[18]</sup> |
| Magnetic Components  | MOTIVELINK (formerly Yongin Electronics) | Korea; manufacturing in India <sup>[25]</sup>  |
| HVDC Contactors      | Kunshan GLVAC Yuantong                   | China; manufacturing in India <sup>[25]</sup>  |

## Vendor Ecosystem

- Technology licensing from Gtake (China), AEM (UK), Landworld (China)
- Local manufacturing partners for magnetic components and contactors
- Sterling Group synergy — fastener manufacturing expertise shared <sup>[12]</sup> <sup>[1]</sup>

## Quality Control Systems

- Testing Facilities: Multiple testing facilities available <sup>[1]</sup>
- Dyno Testing: Up to \_\_\_\_ kW/1200V capability <sup>[1]</sup>
- FAME II Compliance: MCUs compliant or in compliance process <sup>[13]</sup>
- ISO Standards: Implied through ACMA recognition and OEM partnerships

## Automation Systems

- Not explicitly detailed in public sources
- 720K units/year capacity suggests automated assembly lines

## Operational Workflow (Inferred)

1. Technology licensing from partners (Gtake, AEM, Landworld)
2. Component sourcing from global + local vendors
3. Manufacturing at Faridabad plant (MCUs, motors, chargers)
4. Software customization for OEM requirements
5. Vehicle integration testing at Bangalore technical center
6. Validation via dyno testing
7. Delivery to OEMs with application support



# SECTION 6: LEADERSHIP ANALYSIS

## Parent Company Leadership (Sterling Tools Limited)

| Name           | Role                             | Education                                                                                      | Career History                                                                                              | Strategic Contributions                                                                                                      |
|----------------|----------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Anil Aggarwal  | Chairman                         | Not specified                                                                                  | Sterling Group founder/leadership                                                                           | Founded Sterling Tools (1979); strategic EV entry <sup>[26]</sup> <sup>[27]</sup>                                            |
| Atul Aggarwal  | Managing Director                | MBA (Kenan-Flagler); <u>B.Com</u> (SRCC, Delhi); UNC Chapel Hill <sup>[8]</sup> <sup>[9]</sup> | Director at multiple Sterling Group companies; Director at Altigreen Propulsion Labs (2019+) <sup>[8]</sup> | Led EV division creation (2018); MCU business scaling; partnerships with Gtake, AEM, Landworld <sup>[8]</sup> <sup>[9]</sup> |
| Akhil Aggarwal | Whole Time Director              | Not specified                                                                                  | Director since Aug 2019 <sup>[28]</sup>                                                                     | Group operations oversight                                                                                                   |
| Jaideep Wadhwa | Director (SEM Managing Director) | Not publicly detailed                                                                          | Appointed Director Feb 2019 (STL); Sept 2022 (SEM) <sup>[28]</sup> <sup>[2]</sup>                           | MD of SEM; led Bharat Mobility Expo 2025; FY28 systems strategy <sup>[10]</sup> <sup>[19]</sup>                              |
| Pankaj Gupta   | Group CFO                        | Not specified                                                                                  | CFO at Sterling Tools                                                                                       | Financial strategy; Q4 FY25 earnings calls <sup>[5]</sup> <sup>[29]</sup>                                                    |

## SEM Leadership Team (Sterling E-Mobility)

| Name                  | Role                                                |
|-----------------------|-----------------------------------------------------|
| Anuj Sahrawat         | Chief Operations Officer <sup>[1]</sup>             |
| Ramesha TD            | Chief Technology Officer <sup>[1]</sup>             |
| Vishal Sinha          | Head – Business Development & Sales <sup>[1]</sup>  |
| Avinash P Singh       | Head – Quality Assurance <sup>[1]</sup>             |
| Vikram Bishnoi        | Head – HR <sup>[1]</sup>                            |
| Chandrakant Poojari   | Head – Group IT <sup>[1]</sup>                      |
| Himanshu Khanduja     | Head – Sourcing & Vendor Development <sup>[1]</sup> |
| Rupesh Ranjan Prashar | Head – Finance <sup>[1]</sup>                       |

## Key Leadership Insights

- Atul Aggarwal – MBA from top US business school; previously Director at Altigreen Propulsion Labs (EV startup), showing deep EV domain expertise before SEM<sup>[8]</sup>
- Jaideep Wadhwa – Public face of SEM; presented at Bharat Mobility Global Expo 2025; announced FY28 systems strategy<sup>[10]</sup> <sup>[19]</sup>
- Family-led governance – Aggarwal family controls Sterling Group; professional management in SEM operations

# SECTION 7: COMPANY CULTURE

## Company Values

"Entrepreneurial Spirit. Integrity. Team Spirit. Excellence. Sustainable Value."<sup>[1]</sup>

## Work Culture

| Attribute          | Evidence                                                                                |
|--------------------|-----------------------------------------------------------------------------------------|
| Engineering-led    | Mission emphasizes "Engineering led Company" <sup>[1]</sup>                             |
| Customer-centric   | Awards from Ola Electric (delivery), Greaves (trusted partner) <sup>[1]</sup>           |
| Innovation-focused | Technology partnerships; magnet-free motors; localization <sup>[1]</sup> <sup>[7]</sup> |
| Sustainability     | "Sustainable Value" in core values; EV focus <sup>[1]</sup>                             |

## Hiring Strategy

- **Company Size:** 51–200 employees<sup>[24]</sup> <sup>[11]</sup>
- **Focus:** Engineering talent (R&D, application support, QA)
- **Locations:** Faridabad (manufacturing), Bangalore (technical center)

## Employee Growth

- **Rapid scaling** from 2020 incorporation to 51–200 employees by 2024–2025
- **EV business 90% growth** in FY24 suggests hiring acceleration<sup>[15]</sup>

## HR Initiatives

- **Amity Leadership Award** (2023) for control, automation, power engineering development <sup>[1]</sup>
- **Learning ecosystem:** Implied through technical center in Bangalore; engineering focus

## Workplace Reputation

| Award                                               | Year | Significance                                         |
|-----------------------------------------------------|------|------------------------------------------------------|
| OLA Electric Delivery Performance Runner-Up         | 2024 | Customer recognition for reliability <sup>[1]</sup>  |
| Most Trusted Partner – Greaves Electric Mobility    | 2023 | OEM trust <sup>[1]</sup>                             |
| ACMA Excellence Awards Bronze (NPDD & Localization) | 2025 | Industry recognition for localization <sup>[1]</sup> |
| ACMA EV Tech Show Participant                       | 2025 | Industry engagement <sup>[1]</sup>                   |

## Employee Sentiment

- Not publicly available — no Glassdoor/LinkedIn reviews found in search
- Inferred positive: Customer awards, rapid growth, industry recognition suggest stable workplace

# SECTION 8: FINANCIAL ANALYSIS

## Revenue History (Consolidated — Sterling Tools Limited)

| Fiscal Year | Total Revenue      | YoY Growth             | EV Business Revenue                         | EV Growth                       | PAT                                 |
|-------------|--------------------|------------------------|---------------------------------------------|---------------------------------|-------------------------------------|
| FY24        | ~₹770 crore (est.) | —                      | ₹325 crore (EV control units)               | +90%                            | Not specified <sup>[15]</sup>       |
| FY25        | ₹1,038 crore       | +10.6% <sup>[17]</sup> | ~₹435 crore (42% of H1 mix) <sup>[22]</sup> | +56% (9 months) <sup>[16]</sup> | ₹58.3 crore (+5.3%) <sup>[17]</sup> |

## Quarterly Performance (FY25)

| Period        | Revenue                   | EBITDA             | EBITDA Margin | PAT                                |
|---------------|---------------------------|--------------------|---------------|------------------------------------|
| Q3 FY25       | ₹262.7 crore (+12.3% YoY) | ₹31.7 crore (+13%) | 12.1%         | ₹13.6 crore <sup>[16]</sup>        |
| 9 Months FY25 | ₹832.3 crore (+24.5%)     | —                  | —             | ₹49.5 crore (+27%) <sup>[16]</sup> |
| Q2 FY25       | ₹285.9 crore (+35.2%)     | —                  | —             | ₹17.5 crore (+40%) <sup>[22]</sup> |

## SGEM (Sterling Gtake/E-Mobility) Specifics

| Metric                      | Value                                   |
|-----------------------------|-----------------------------------------|
| 9-Month FY25 Revenue        | ₹345.9 crore (+56% YoY) <sup>[16]</sup> |
| Q4 FY22 Monthly Run Rate    | ₹100 million <sup>[13]</sup>            |
| Q4 FY22 Revenue             | ₹309 million <sup>[13]</sup>            |
| EV Business Share (H1 FY25) | 42% of total mix <sup>[22]</sup>        |
| EV Business Share (H1 FY24) | 30% of total mix <sup>[22]</sup>        |

## Financial Structure

| Parameter                | Value                              |
|--------------------------|------------------------------------|
| Authorized Share Capital | ₹30,00,00,000 <sup>[2]</sup>       |
| Paid-up Capital          | ₹28,45,60,000 <sup>[2]</sup>       |
| EBITDA Margin (FY25)     | 12.8% consolidated <sup>[17]</sup> |
| PAT Margin (FY25)        | 5.6% consolidated <sup>[17]</sup>  |

| Parameter       | Value                         |
|-----------------|-------------------------------|
| Dividend (FY25) | 200% declared <sup>[17]</sup> |

### Investments & Expansion Spending

- Faridabad EV campus expansion for OBC/DC-DC manufacturing<sup>[18]</sup>
- Technology licensing fees paid to Gtake, AEM, Landworld
- Bangalore technical center setup for R&D<sup>[23]</sup>

### Debt & Assets

- Not publicly disclosed in detail for SEM subsidiary
- Parent company appears low-debt (strong cash flow from 200% dividend)

### Growth Trends

| Trend             | Observation                                                   |
|-------------------|---------------------------------------------------------------|
| EV Revenue Growth | +90% in FY24; +56% in 9M FY25 <sup>[15]</sup> <sup>[16]</sup> |
| Revenue Mix Shift | EV 30% (H1 FY24) → 42% (H1 FY25) <sup>[22]</sup>              |
| Systems Target    | 70% integrated systems by FY28 <sup>[19]</sup>                |
| Capacity          | 720K MCUs/year (expandable) <sup>[1]</sup>                    |

## SECTION 9: MARKET POSITION

### Main Competitors in India EV Components

| Competitor                 | Product Focus               | Strengths                            | Weaknesses                       |
|----------------------------|-----------------------------|--------------------------------------|----------------------------------|
| Bosch India                | EV powertrain, MCUs, motors | Global technology; OEM relationships | Higher cost; slower localization |
| Motherson Sumi             | EV components, wiring       | Scale; OEM partnerships              | Less MCU focus                   |
| Sona BLW (Sona Comstar)    | EV motors, differential     | Strong EV portfolio; global          | Less MCU presence                |
| Tata Automotive Components | EV parts                    | Tata group synergy                   | Internal focus                   |
| Yamaha Motor India         | EV components               | Motor expertise                      | Limited to own brand             |
| Indian MCUs Startups       | MCUs                        | Agility                              | Limited scale                    |

**Note:** Sterling E-Mobility is **India's largest MCU manufacturer** with 50%+ market share — no domestic competitor matches this scale in MCUs specifically<sup>[1]</sup>

## Comparison Against Major EV OEMs (Not Direct Competitors)

| Company           | Relationship to SEM                                                 |
|-------------------|---------------------------------------------------------------------|
| Tata EV           | Potential/Existing OEM customer                                     |
| Ola Electric      | <b>Largest customer</b> (until Gen3 in-house shift) <sup>[17]</sup> |
| Ather Energy      | Potential OEM customer                                              |
| Mahindra Electric | Potential OEM customer                                              |
| Hero Electric     | Potential OEM customer                                              |

**Clarification:** SEM is a **component supplier**, not an EV manufacturer. It does not compete with Tata EV, Ola Electric, Ather, etc. — it supplies them.

## SWOT Analysis

| Strengths                                                           | Weaknesses                                                         |
|---------------------------------------------------------------------|--------------------------------------------------------------------|
| India's largest MCU manufacturer (50%+ market share) <sup>[1]</sup> | Customer concentration (Ola Electric dependency) <sup>[17]</sup>   |
| 720K units/year capacity with scalability <sup>[1]</sup>            | Technology licensing dependence (China/UK partners) <sup>[1]</sup> |
| 15+ OEM customers including Ola, Greaves <sup>[14]</sup>            | Limited brand recognition outside B2B                              |
| ACMA award for localization <sup>[1]</sup>                          | No proprietary MCU technology (licensed from Gtake)                |
| Integrated powertrain expansion (2-in-1, 3-in-1) <sup>[12]</sup>    | Small team (51–200 employees) <sup>[11]</sup>                      |
| Strong parent company (Sterling Tools, 40+ years) <sup>[1]</sup>    |                                                                    |
| Opportunities                                                       | Risks/Threats                                                      |
| -----                                                               | -----                                                              |
| India EV market growing rapidly (FAME II, PLI schemes)              | OEMs moving to in-house MCUs (like Ola Gen3) <sup>[17]</sup>       |
| 70% integrated systems target by FY28 <sup>[19]</sup>               | China-India geopolitical tensions affecting tech partnerships      |
| Magnet-free motors (no rare-earth dependency) <sup>[7]</sup>        | Competition from global players (Bosch, Continental)               |
| LCV, commercial EV segment expansion <sup>[13]</sup>                | Raw material price volatility                                      |
| Export potential to Southeast Asia, Europe                          | Regulatory changes in EV subsidies                                 |
| 800V architecture adoption (high-voltage OBCs) <sup>[12]</sup>      | Customer churn if relationships weaken                             |

# SECTION 10: EV INDUSTRY POSITIONING

## Role in India's EV Ecosystem

| Aspect                      | SEM's Position                                                                |
|-----------------------------|-------------------------------------------------------------------------------|
| Critical Component Supplier | MCUs are "brains" of EVs — essential for all electric vehicles <sup>[1]</sup> |
| Import Substitution Leader  | Localization reduces China dependency for MCUs <sup>[1]</sup> <sup>[25]</sup> |

| Aspect          | SEM's Position                                                                 |
|-----------------|--------------------------------------------------------------------------------|
| FAME II Enabler | FAME II-compliant MCUs support government EV push <sup>[13]</sup>              |
| OEM Enabler     | 15+ OEMs depend on SEM for MCUs — critical supply chain player <sup>[14]</sup> |

Strategic Importance

- **National Security:** Domestic MCU production reduces critical component dependency on China
- **Employment:** Engineering jobs in Faridabad, Bangalore
- **Technology Absorption:** License-based model enables Indian manufacturing capability

Market Influence

- 50%+ MCU market share — pricing power, OEM influence<sup>[1]</sup>
- **Trendsetter:** Magnet-free motors could redefine motor technology in India<sup>[7]</sup>

Industry Recognition

| Award/Recognition                                   | Year | Issuer                                   |
|-----------------------------------------------------|------|------------------------------------------|
| ACMA Excellence Awards Bronze (NPDD & Localization) | 2025 | ACMA <sup>[1]</sup>                      |
| OLA Electric Delivery Performance Runner-Up         | 2024 | Ola Electric <sup>[1]</sup>              |
| Most Trusted Partner                                | 2023 | Greaves Electric Mobility <sup>[1]</sup> |
| Amity Leadership Award                              | 2023 | Amity University <sup>[1]</sup>          |

Government-Linked Initiatives

- **FAME II Compliance:** MCUs compliant with government EV subsidy requirements<sup>[13]</sup>
- **NPDD (Non-Publicly Disclosed) Localization:** ACMA award suggests alignment with government localization push<sup>[1]</sup>
- **PLI Schemes:** Likely beneficiary (EV components) though not explicitly confirmed

SECTION 11: PARTNERSHIPS AND COLLABORATIONS

Technology Partnerships

| Partner                          | Country          | Technology                 | Impact                                                                                                  |
|----------------------------------|------------------|----------------------------|---------------------------------------------------------------------------------------------------------|
| Jiangsu Gtake Electric Co. Ltd.  | China (Shenzhen) | Motor Control Units (MCUs) | Core MCU technology; exclusive Indian rights; multiple products localized <sup>[1]</sup> <sup>[7]</sup> |
| Advanced Electric Machines (AEM) | UK (Washington)  | Magnet-Free Motors         | Eliminates rare-earth magnet dependency; cost + supply chain advantage <sup>[1]</sup> <sup>[7]</sup>    |

| Partner                         | Country          | Technology                                | Impact                                                                                   |
|---------------------------------|------------------|-------------------------------------------|------------------------------------------------------------------------------------------|
| Landworld Technology Co. Ltd.   | China (Shenzhen) | On-Board Chargers, DC-DC Converters, PDUs | Entered charger segment (2025); 3-in-1 integrated systems <sup>[1]</sup> <sup>[18]</sup> |
| MOTIVELINK (Yongin Electronics) | Korea            | Magnetic Components for EVs               | Manufacturing in India; supply chain localization <sup>[25]</sup>                        |
| Kunshan GLVAC Yuantong          | China            | HVDC Contactors & Relays                  | Manufacturing in India; localization <sup>[25]</sup>                                     |

### OEM Partnerships

| OEM                       | Segment           | Relationship Status                                                                              |
|---------------------------|-------------------|--------------------------------------------------------------------------------------------------|
| Ola Electric              | 2W (scooters)     | Largest customer (until Gen3 in-house shift); delivery award 2024 <sup>[17]</sup> <sup>[1]</sup> |
| Greaves Electric Mobility | 3W (E-CNGON)      | "Most Trusted Partner" award 2023 <sup>[1]</sup>                                                 |
| 15+ EV OEMs               | 2W, 3W, LCV, PCV  | Active supply relationships <sup>[14]</sup>                                                      |
| 30 additional OEMs        | Multiple segments | Advanced discussions for MCU requirements <sup>[13]</sup>                                        |

### Academic Collaborations

- Amity University: Leadership Award 2023 for control, automation, power engineering development <sup>[1]</sup>.

### Manufacturing Partners

- Faridabad EV campus: SEM's own facility (no third-party manufacturing partners mentioned)

### Impact of Partnerships

| Partnership       | Strategic Impact                                                         |
|-------------------|--------------------------------------------------------------------------|
| Gtake (China)     | Enabled rapid MCU market entry; India's largest MCU manufacturer         |
| AEM (UK)          | Magnet-free motors differentiate from competitors; supply chain security |
| Landworld (China) | Expanded into OBC/DC-DC; 3-in-1 systems for FY28 target                  |
| Ola Electric      | Scaled production to 720K capacity; 90% EV growth in FY24                |

## SECTION 12: FUTURE ROADMAP

## Announced Strategic Goals

| Timeline  | Goal                                                              | Source          |
|-----------|-------------------------------------------------------------------|-----------------|
| FY2028    | 70%+ portfolio as integrated powertrain/power electronics systems | <sup>[19]</sup> |
| FY2027    | New customer programs go into production; pivot results visible   | <sup>[19]</sup> |
| 2025–2028 | Ramp up traction motors, controllers, multi-function drive units  | <sup>[19]</sup> |

## Future Projects (Evidence-Based)

| Project                       | Description                                  | Likelihood                                                        |
|-------------------------------|----------------------------------------------|-------------------------------------------------------------------|
| 3-in-1 System Mass Production | OBC + DC-DC + PDU integrated unit            | High (already announced) <sup>[12]</sup>                          |
| Magnet-Free Motor Scale-up    | AEM UK technology commercialization          | High (partnership active) <sup>[7]</sup>                          |
| 800V Architecture Products    | High-voltage MCUs/OBCs for fast-charging EVs | Medium (22kW OBC up to 750Vdc implies capability) <sup>[12]</sup> |
| Export Expansion              | Southeast Asia, Europe OEMs                  | Medium (parent company global presence) <sup>[12]</sup>           |
| New Manufacturing Line        | Additional capacity beyond 720K MCUs         | Medium (expandable capacity stated) <sup>[1]</sup>                |

## Expansion Plans

| Area              | Plan                                                       |
|-------------------|------------------------------------------------------------|
| Faridabad Plant   | OBC/DC-DC production at EV campus <sup>[18]</sup>          |
| Product Portfolio | From MCU-only to full powertrain systems <sup>[1]</sup>    |
| Customer Base     | 30 additional OEMs in advanced discussions <sup>[13]</sup> |

## Planned Investments

- Technology licensing fees (ongoing to Gtake, AEM, Landworld)
- Capital expenditure for Faridabad expansion (not quantified publicly)
- R&D spending at Bangalore technical center (not quantified)

## Growth Projections (Evidence-Based Forecast)

| Year | Projected Revenue (SEM) | Key Drivers                                                       |
|------|-------------------------|-------------------------------------------------------------------|
| 2026 | ₹500–600 crore          | Existing OEMs + 30 new OEMs; OBC/DC-DC ramp-up                    |
| 2027 | ₹700–850 crore          | FY28 systems target begins; new customer programs <sup>[19]</sup> |
| 2028 | ₹1,000+ crore           | 70% systems portfolio; magnet-free motors scale                   |
| 2030 | ₹1,500+ crore (est.)    | India EV market maturity; export markets                          |



Assumptions:

- India EV CAGR ~30–40% through 2030
- SEM maintains 40–50% MCU market share
- Systems transition successful (70% target)
- No major customer churn (Ola dependency mitigated)

SECTION 13: RISKS

Risk Assessment (Ranked by Severity)

| Risk Category            | Specific Risk                                                 | Level       | Mitigation                                                                       |
|--------------------------|---------------------------------------------------------------|-------------|----------------------------------------------------------------------------------|
| Customer Concentration   | Ola Electric = largest customer; churn risk (Gen3 in-house)   | HIGH        | 30 OEMs in discussions; diversification strategy <sup>[17]</sup> <sup>[13]</sup> |
| Technology Dependency    | MCU tech licensed from Gtake (China); no proprietary IP       | HIGH        | AEM UK partnership; localization; building in-house capability                   |
| Geopolitical             | China-India tensions affecting Gtake/Landworld partnerships   | MEDIUM-HIGH | AEM UK partnership diversifies; localization efforts <sup>[2]</sup>              |
| Competition              | Global players (Bosch, Continental) entering India MCU market | MEDIUM      | 50% market share; first-mover advantage; cost advantage <sup>[1]</sup>           |
| OEM Vertical Integration | More OEMs making in-house MCUs (like Ola)                     | MEDIUM-HIGH | Focus on SMEs/startups lacking R&D budget; systems products                      |
| Supply Chain             | Raw material (semiconductors, magnets) volatility             | MEDIUM      | MOTIVELINK, GLVAC localization; vendor diversification <sup>[25]</sup>           |
| Regulatory               | FAME II subsidy cuts; EV policy changes                       | MEDIUM      | FAME II-compliant products; export markets                                       |
| Technology Obsolescence  | 800V architectures; new motor tech                            | MEDIUM      | 22kW OBC up to 750Vdc; AEM magnet-free motors <sup>[12]</sup> <sup>[2]</sup>     |
| Financial                | Revenue slowdown (Q4 FY25 Ola decline)                        | MEDIUM      | Diversification; 70% systems target for higher margins <sup>[17]</sup>           |
| Operational              | Small team (51–200 employees) scaling challenges              | LOW-MEDIUM  | Parent company support; Faridabad/Bangalore locations                            |

SECTION 14: INTERVIEW INTELLIGENCE

Company Values (From Official Source)

"Entrepreneurial Spirit. Integrity. Team Spirit. Excellence. Sustainable Value."<sup>[1]</sup>

## Likely Interview Themes

| Category   | Focus Areas                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------|
| Technical  | EV powertrain, MCU architecture, BLDC/PMSM motors, power electronics, embedded software, CAN-BUS, UDS |
| Domain     | FAME II norms, EV standards, automotive safety (ISO 26262), vehicle integration                       |
| Behavioral | Customer-centricity, entrepreneurial mindset, problem-solving, teamwork                               |
| Strategic  | India EV market, localization, supply chain, competition, FY28 systems goal                           |

### Technical Focus Areas

- **Motor Control:** Field-oriented control (FOC), PWM, torque control, speed regulation
- **Power Electronics:** DC-DC converters, OBC topologies, SiC/GaN devices
- **Embedded Systems:** ARM Cortex MCUs, real-time OS, automotive Ethernet
- **Testing:** Dyno testing, HIL (Hardware-in-Loop), validation protocols
- **Software:** UDS (Unified Diagnostic Services), CAN-BUS communication, calibration

### HR Expectations

- **Alignment with values:** Entrepreneurial spirit, integrity, excellence
- **Customer focus:** Experience with OEMs, delivery accountability
- **Adaptability:** Rapidly evolving EV industry, new technologies
- **Learning mindset:** New product launches (OBC, DC-DC, magnet-free motors)

### Leadership Expectations

- **Jaideep Wadhwa (MD):** Strategic thinker, EV domain expertise, OEM relationship management
- **Atul Aggarwal (MD, Parent):** Long-term vision, technology partnerships, localization
- **Anuj Sahrawat (COO):** Operational excellence, manufacturing efficiency
- **Ramesha TD (CTO):** Technical innovation, R&D, new product development

## Top 50 Likely Interview Questions

### Company & Industry Knowledge (10 Questions)

1. What is Sterling E-Mobility's core business?
2. Who are SEM's main technology partners?
3. What is our market share in India's MCU segment?
4. Explain the difference between BLDC, PMSM, and Induction motors.
5. What is FAME II, and how does it affect SEM's products?

6. What is the FY28 strategic goal announced by SEM?
7. Why are magnet-free motors strategically important?
8. Name 3 OEMs SEM supplies to.
9. What is a 3-in-1 powertrain system?
10. How does SEM's business model differ from Tata EV or Ola Electric?

### **Technical — MCU & Motor Control (10 Questions)**

11. Explain the working principle of a Motor Control Unit.
12. What is Field-Oriented Control (FOC)?
13. How do you calculate torque in a PMSM motor?
14. What is the role of PWM in motor control?
15. Explain Hall sensor vs. Resolver vs. Encoder.
16. What is the maximum voltage range of SEM's MCUs?
17. How do you protect an MCU from overcurrent?
18. What is SiC, and why is it used in EV power electronics?
19. Explain the difference between low-voltage and high-voltage EV architectures.
20. What is ISO 26262, and what ASIL level is required for MCUs?

### **Technical — Power Electronics (8 Questions)**

21. Explain the working of an On-Board Charger (OBC).
22. What is the efficiency of SEM's 22kW OBC?
23. Draw the topology of a DC-DC converter.
24. What is the difference between buck and boost converters?
25. Why is liquid cooling used in high-power OBCs?
26. What is a PDU (Power Distribution Unit)?
27. Explain the 2-in-1 and 3-in-1 integrated systems.
28. What is the voltage range of SEM's 22kW OBC?

### **Technical — Software & Embedded (7 Questions)**

29. What is UDS (Unified Diagnostic Services)?
30. Explain CAN-BUS communication protocol.
31. What is AUTOSAR, and how is it used in EVs?
32. What embedded OS do you have experience with?
33. How do you debug communication issues in CAN-BUS?
34. What is calibration in motor control?

35. Explain the difference between hardware and software testing.

### Behavioral & Situational (8 Questions)

- 36. Describe a time you solved a critical customer issue under pressure.
- 37. How do you handle conflicting priorities between R&D and delivery?
- 38. Give an example of when you showed entrepreneurial initiative.
- 39. Describe a situation where you worked in a cross-functional team.
- 40. How do you ensure quality while meeting tight deadlines?
- 41. Tell us about a time you failed and what you learned.
- 42. How do you stay updated with EV industry trends?
- 43. Why do you want to work at Sterling E-Mobility?

### Strategic & Career (5 Questions)

- 44. Where do you see India's EV industry in 2030?
- 45. How can SEM reduce dependency on Chinese technology partners?
- 46. What is your understanding of SEM's FY28 systems strategy?
- 47. Where do you see yourself in 5 years?
- 48. What skills do you want to develop at SEM?

### Company-Specific (5 Questions)

- 49. What are SEM's core values?
- 50. How does SEM's Bangalore technical center support Faridabad manufacturing?

## SECTION 15: STOCK MARKET & INVESTOR INTELLIGENCE

### Listing Status

| Attribute           | Status                                                                 |
|---------------------|------------------------------------------------------------------------|
| Is SEM Listed?      | NO — Privately held subsidiary <sup>[9]</sup> <sup>[2]</sup>           |
| Parent Company      | Sterling Tools Limited (STL)                                           |
| Parent Stock Symbol | STERTOOLS (BSE: 532404, NSE: STERTOOLS) <sup>[5]</sup> <sup>[17]</sup> |
| Parent Exchange     | BSE & NSE (India) <sup>[15]</sup>                                      |
| SEM Ownership       | 100% subsidiary of Sterling Tools Limited <sup>[14]</sup>              |

## Parent Company (Sterling Tools) Stock Performance

| Metric                | Value                                                   |
|-----------------------|---------------------------------------------------------|
| FY25 Revenue          | ₹1,038 crore (+10.6% YoY) <sup>[17]</sup>               |
| FY25 PAT              | ₹58.3 crore (+5.3% YoY) <sup>[17]</sup>                 |
| EBITDA Margin         | 12.8% consolidated <sup>[17]</sup>                      |
| PAT Margin            | 5.6% consolidated <sup>[17]</sup>                       |
| Dividend (FY25)       | 200% declared <sup>[17]</sup>                           |
| Stock Return (Recent) | +30% in revenue; 90% EV business growth <sup>[15]</sup> |

## Major Investors (Parent Company)

| Investor Type | Details                                                                      |
|---------------|------------------------------------------------------------------------------|
| Promoters     | Aggarwal Family (Anil, Atul, Akhill Agarwal) <sup>[26]</sup> <sup>[27]</sup> |
| Institutional | Not detailed in search results                                               |
| Retail        | Publicly traded; retail investors on BSE/NSE                                 |

## Investor-Style Analysis

### Bull Case for SEM (via SterlingTools):

- India's largest MCU manufacturer with 50%+ market share <sup>[11]</sup>
- EV business grew 90% in FY24; 56% in 9M FY25 <sup>[16]</sup> <sup>[15]</sup>
- EV revenue mix: 30% → 42% in one year <sup>[22]</sup>
- 70% systems target by FY28 = higher margins <sup>[19]</sup>
- Magnet-free motors = competitive differentiation <sup>[7]</sup>
- 30 OEMs in advanced discussions = pipeline <sup>[13]</sup>

### Bear Case:

- Ola Electric churn risk (Gen3 in-house MCU) <sup>[17]</sup>
- Technology licensing (no proprietary MCU IP) <sup>[11]</sup>
- China partnerships (geopolitical risk) <sup>[7]</sup>
- Q4 FY25 slowdown already visible <sup>[17]</sup>

### Valuation Context:

- Parent company market cap: Not available in search
- EV segment trading premium vs. traditional auto components
- SEM likely valued at 15–25x P/E by 2028 if systems target achieved

# SECTION 16: COMPLETE MASTER TIMELINE

| Date         | Event                                                                     | Business Impact                                                        | Long-Term Impact                             |
|--------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------|
| 1979         | Sterling Tools Limited incorporated                                       | Parent company founded; 40+ years automotive expertise                 | Foundation for EV venture                    |
| 2018         | Sterling Tools establishes E-Mobility Division                            | Strategic EV entry 2 years before subsidiary <sup>[11]</sup>           | First-mover advantage in India EV components |
| Jan 10, 2020 | SEM incorporated as "Sterling Gtake E-Mobility Ltd"                       | Legal entity for EV business; CIN U31909DL2020PLC360123 <sup>[2]</sup> | Corporate structure for scaling              |
| 2020         | MCU manufacturing begins with Gtake license                               | First EV product launched; FAME II era entry <sup>[1]</sup>            | India's domestic MCU capability              |
| 2021         | 300,000+ MCUs manufactured; 15+ OEMs supplied                             | Scale achieved in Year 1 <sup>[14]</sup>                               | Market leadership established                |
| FY2022 Q4    | Revenue ₹309M; monthly run rate ₹100M                                     | Profitable operations validated <sup>[13]</sup>                        | Sustainable business model confirmed         |
| FY2023       | Expanded to 3W, motorcycle, LCV customers                                 | Product diversification <sup>[13]</sup>                                | Reduced 2W dependency                        |
| 2023         | "Most Trusted Partner" award from Greaves Electric Mobility               | OEM validation <sup>[1]</sup>                                          | Reputation building                          |
| 2023         | Amity Leadership Award for control/automation engineering                 | Industry recognition <sup>[1]</sup>                                    | Talent attraction                            |
| FY2024       | EV business +90%; EV control unit revenue ₹325 crore                      | Major growth milestone <sup>[15]</sup>                                 | EV became core business                      |
| 2024         | OTA Electric "Delivery Performance Runner-Up" award                       | Key customer recognition <sup>[1]</sup>                                | Quality reputation                           |
| H1 FY25      | EV revenue mix 42% (up from 30% in H1 FY24)                               | EV now dominant segment <sup>[22]</sup>                                | Strategic pivot complete                     |
| 9M FY25      | SGEM revenue ₹345.9 crore (+56% YoY); consolidated PAT ₹49.5 crore (+27%) | Sustained growth <sup>[16]</sup>                                       | Financial strength                           |
| Q2 FY25      | Consolidated PAT ₹17.5 crore (+40%); revenue ₹285.9 crore (+35.2%)        | Strong quarterly performance <sup>[22]</sup>                           | Investor confidence                          |
| 2025         | Partnership with Advanced Electric Machines (UK) for magnet-free motors   | Technology diversification <sup>[7]</sup>                              | Supply chain security                        |
| 2025         | Partnership with Landworld Technology (China) for OBC/DC-DC               | Entered charger segment <sup>[18]</sup>                                | Product portfolio expanded                   |
| Q4 FY25      | Slowdown due to Ola Electric Gen3 in-house MCU shift                      | Revenue pressure <sup>[17]</sup>                                       | Customer concentration risk exposed          |
| FY2025       | Consolidated revenue ₹1,038 crore (+10.6%); PAT ₹58.3 crore (+5.3%)       | Overall growth despite Q4 slowdown <sup>[17]</sup>                     | Resilience demonstrated                      |
| FY2025       | 200% dividend declared                                                    | Strong cash flow <sup>[17]</sup>                                       | Shareholder returns                          |

| Date      | Event                                                   | Business Impact                                           | Long-Term Impact                |
|-----------|---------------------------------------------------------|-----------------------------------------------------------|---------------------------------|
| Oct 2025  | Rebranded to "Sterling E-Mobility Solutions Limited"    | Reflects expanded portfolio <sup>[5]</sup> <sup>[7]</sup> | Strategic messaging updated     |
| 2025      | ACMA Excellence Awards Bronze (NPDD & Localization)     | Industry recognition <sup>[1]</sup>                       | Government alignment            |
| 2025      | Installed capacity 720K MCUs/year; 50%+ market share    | Market leadership confirmed <sup>[1]</sup>                | Competitive moat                |
| Dec 2025  | Announced FY28 target: 70% integrated systems portfolio | Strategic roadmap clarified <sup>[19]</sup>               | Future direction set            |
| 2026      | New customer programs expected to go into production    | FY27 results visible <sup>[19]</sup>                      | Growth reacceleration           |
| 2027–2028 | 70% systems portfolio target; magnet-free motors scale  | FY28 strategic goal                                       | Higher margins, differentiation |
| 2030      | (Projected) ₹1,500+ crore revenue; export markets       | Long-term vision                                          | India EV market leader          |

## SECTION 17: EXECUTIVE SUMMARY

### PAST (2018–2023)

- **2018:** Sterling Tools created E-Mobility Division, anticipating India's EV transition
- **2020:** Incorporated Sterling Gtake E-Mobility with Jiangsu Gtake (China) MCU technology license
- **2021–2023:** Rapid scale — 300K+ MCUs, 15+ OEMs, FAME II compliance, industry awards
- **Key achievement:** Became India's largest MCU manufacturer within 2 years

### PRESENT (2024–2025)

- **Market position:** 50%+ India MCU market share; 720K units/year capacity
- **Revenue:** EV business ₹325 crore (FY24); 42% of parent company mix (H1 FY25)
- **Growth:** 90% EV growth (FY24); 56% (9M FY25)
- **Products:** MCUs, motors, OBC (6.6–22kW), DC-DC converters, 2-in-1/3-in-1 systems
- **Partnerships:** Gtake (China, MCU), AEM (UK, magnet-free motors), Landworld (China, OBC)
- **Challenge:** Ola Electric churn (Gen3 in-house); Q4 FY25 slowdown
- **Recognition:** ACMA Bronze Award 2025; Ola delivery award 2024; Greaves trusted partner 2023

## FUTURE (2026–2030)

- **FY2028 Goal:** 70%+ portfolio as integrated powertrain systems (from ~30% today)
- **2026–2027:** 30 new OEMs in pipeline; new customer programs production-ready
- **Technology:** Magnet-free motors scale-up; 800V architecture products
- **Revenue Projection:** ₹1,000+ crore by FY28; ₹1,500+ crore by 2030 (est.)
- **Strategic priorities:**
  - Mitigate Ola dependency (diversification)
  - Build proprietary capability (reduce China licensing reliance)
  - Export markets (Southeast Asia, Europe)
  - Systems leadership (higher margins)

## INVESTMENT THESIS

**BULL:** India's largest MCU player, 90% EV growth, 70% systems target = margin expansion, magnet-free motors = differentiation, parent company strong (200% dividend)

**BEAR:** Ola churn exposed customer concentration, no proprietary MCU IP, China geopolitical risk, Q4 FY25 slowdown already visible

**VERDICT:** High-growth B2B EV component leader with execution track record, but customer concentration and technology dependency are key risks to monitor.

## SOURCE CONFIDENCE LEVEL

| Information Type                       | Confidence | Reason                                                                                  |
|----------------------------------------|------------|-----------------------------------------------------------------------------------------|
| Incorporation date (Jan 10, 2020)      | HIGH       | MCA official records <sup>[2]</sup> <sup>[4]</sup>                                      |
| Market share (50%+ MCU)                | HIGH       | Company website <sup>[1]</sup>                                                          |
| Revenue figures (FY24, FY25)           | HIGH       | NSE filings, ET Auto, Business Standard <sup>[15]</sup> <sup>[17]</sup> <sup>[16]</sup> |
| Capacity (720K units/year)             | HIGH       | Company website <sup>[1]</sup>                                                          |
| Technology partnerships                | HIGH       | Company website, press releases <sup>[1]</sup> <sup>[7]</sup> <sup>[18]</sup>           |
| Leadership details                     | HIGH       | MCA, company website, NSE filings <sup>[2]</sup> <sup>[1]</sup> <sup>[8]</sup>          |
| Product specifications                 | HIGH       | Company website product pages <sup>[12]</sup> <sup>[21]</sup>                           |
| FY28 strategy (70% systems)            | HIGH       | ET Manufacturing interview <sup>[19]</sup>                                              |
| Ola Electric churn impact              | HIGH       | NSE filing, ET Auto <sup>[17]</sup>                                                     |
| Employee count (51–200)                | MEDIUM     | LinkedIn <sup>[11]</sup> <sup>[24]</sup>                                                |
| 2030 revenue projection (₹1,500 crore) | LOW-MEDIUM | Analyst estimate based on EV CAGR assumptions                                           |
| Interview questions (top 50)           | LOW-MEDIUM | Inferred from company profile, industry standards                                       |



**Report Prepared By: Corporate Intelligence Research Analyst**

**Date: June 2, 2026**

**Coverage: Sterling E-Mobility Solutions Limited (formerly Sterling Gtake E-Mobility) — 100% subsidiary of Sterling Tools Limited (BSE: 532404, NSE: STERTOOLS)**

**\*\***

1. <https://www.sterlingemobility.com/company>
2. <https://www.indiafilings.com/search/sterling-emobility-solutions-limited-cin-U31909DL2020PLC360123>
3. <https://in.linkedin.com/company/sterlinggtake>
4. <https://www.indiafilings.com/search/sterling-gtake-emobility-limited-cin-U31909DL2020PLC360123>
5. <https://stlfasteners.com/assets/upload/investors/20251106152119-press-release-re-brand-178002696712.pdf>
6. <https://themachinemaker.com/news/sterling-gtake-renamed-as-sterling-e-mobility-solutions-limited-to-strengthen-focus-on-ev-technologies/>
7. <https://manufacturing.economictimes.indiatimes.com/news/industry/sterling-e-mobility-solutions-rebrands-to-enhance-ev-technology-offerings/124549045>
8. <https://in.marketscreener.com/insider/ATUL-AGGARWAL-A0UGYX/>
9. <https://www.youtube.com/watch?v=Nyw9ZQHUZ-4>
10. <https://www.facebook.com/EconomicTimesAuto/videos/jaideep-wadhwa-managing-director-sterling-gtake-e-mobility-highlights-how-the-bh/912302560889569/>
11. <https://in.linkedin.com/company/sterlingemobilitysolutions>
12. <https://www.sterlingemobility.com/product-category/power-electronics/high-voltage>
13. <https://www.businesswireindia.com/sterling-gtake-e-mobility-ltd-is-set-to-take-a-quantum-leap-in-fy23-79059.html>
14. <https://www.electronicstheory.biz/eb-specials/sterling-to-make-more-ev-components/>
15. <https://www.entrepreneur.com/en-in/news-and-trends/sterling-tools-records-90-growth-in-ev-business/476947>
16. <https://www.mobilityoutlook.com/news/sterling-tools-reports-27-growth-in-profit-for-nine-months-of-fy25/>
17. <https://auto.economictimes.indiatimes.com/news/auto-components/sterling-tools-reports-10-6-rise-in-fy25-total-income/121156980>
18. <https://www.sterlingemobility.com/news/sterling-e-mobility-solutions-limited-partners-with-china-s-landworld-to-locally-manufacture-ev-on-board-chargers-and-dc-dc-converters>
19. <https://manufacturing.economictimes.indiatimes.com/news/automotive/sterling-e-mobility-to-make-systems-core-of-business-targets-70-share-by-fy28/125888784>
20. <https://www.icra.in/Rating/GetRationalReportFilePdf?id=138866>
21. <https://sterlinggtake.com/motor-controller-products/>
22. <https://money.rediff.com/news/market/sterling-tools-profit-soars-40-in-q2-fy24/18502920241115>
23. <https://www.sterlingemobility.com/contact-us/d04>
24. <https://www.zoominfo.com/c/sterling-gtake-e-mobility-ltd/547019494>
25. <https://forum.valuepickr.com/t/sterling-tools-stl-fasteners-ev-mcus/21747?page=4>
26. <https://www.business-standard.com/companies/quarterly-results/sterling-tools-ltd-quarterly-results-5110>
27. <https://www.moneycontrol.com/company-facts/sterlingtools/management/STI>

28. <https://stlfasteners.com/assets/upload/investors/20240111122058-2-729141295393.pdf>
29. <https://stlfasteners.com/assets/upload/investors/20250521115044-stltrascriptq4fy25-525506397546.pdf>
30. [https://www.linkedin.com/posts/sterlingmobilitysolutions\\_sterlingtake-landworld-ev-activity-7374012104502149120-QHDA](https://www.linkedin.com/posts/sterlingmobilitysolutions_sterlingtake-landworld-ev-activity-7374012104502149120-QHDA)
31. <https://www.st.com/en/automotive-microcontrollers/stellar-e-electrification.html>
32. [https://nsearchives.nseindia.com/corporate/STERTOOLS\\_13052024181459\\_recording.pdf](https://nsearchives.nseindia.com/corporate/STERTOOLS_13052024181459_recording.pdf)
33. <https://stlfasteners.com/our-leadership>
34. <https://www.zoominfo.com/p/Jaideep-Wadhwa/1483119598>
35. [https://cdn.olaelectric.com/sites/evdp/pages/investor/lodr/investor\\_relations\\_contact\\_v1.pdf](https://cdn.olaelectric.com/sites/evdp/pages/investor/lodr/investor_relations_contact_v1.pdf)
36. <https://themachinemaker.com/news/sterling-tools-appoints-atul-aggarwal-as-managing-director/>
37. <https://stlfasteners.com/sem>
38. [https://nsearchives.nseindia.com/corporate/STERTOOLS\\_02092019143504\\_NOTICEAGM2019\\_032.PDF](https://nsearchives.nseindia.com/corporate/STERTOOLS_02092019143504_NOTICEAGM2019_032.PDF)